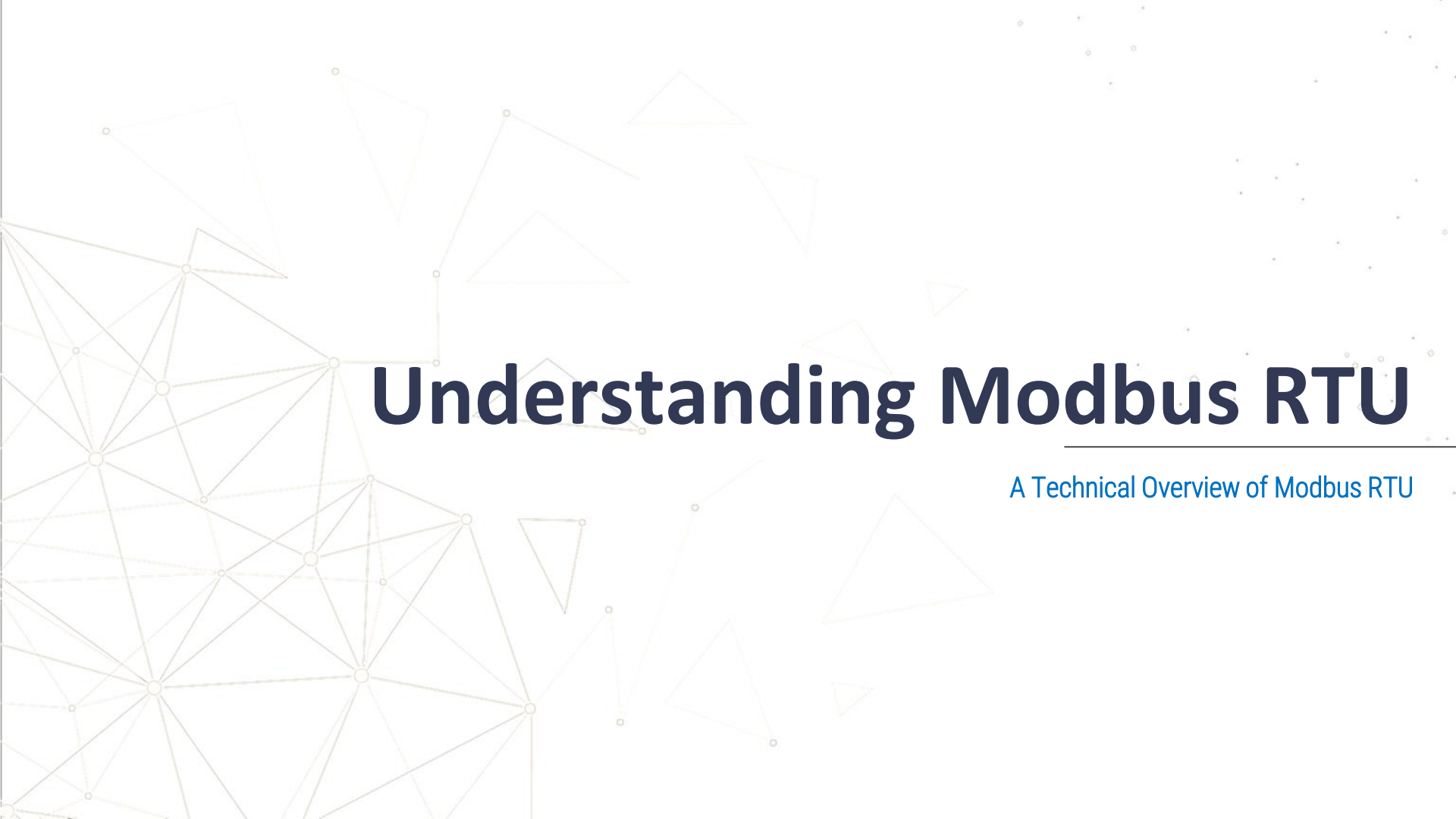


The background features a complex network of thin, light-colored lines connecting small circular nodes, primarily concentrated on the left side. Scattered across the entire background are various geometric shapes, including triangles and polygons, some of which are outlined in a light yellow or green color. The overall aesthetic is technical and modern.

Understanding Modbus RTU

A Technical Overview of Modbus RTU

Understanding Modbus RTU

- Introduction to Modbus RTU
- Modbus RTU Message Structure: PDU & ADU
- Modbus RTU Message Framing
- Modbus RTU Master-Slave Message Flow
- CRC in Modbus RTU: Ensuring Data Integrity
- Timing in Modbus RTU Communication

Introduction to Modbus RTU

Modbus RTU Basics

- A serial communication protocol commonly used in industrial automation for its efficiency and reliability.
- Operates with a master-slave model, typically uses RS232 or RS485 serial interfaces.

Why Modbus RTU

- **Efficiency:** Uses a compact binary message format for fast communication and effective bandwidth use.
- **Reliability:** Incorporates CRC error checking, ensuring data integrity.
- **Wide Applicability:** Utilized in areas like industrial automation, process control, and energy monitoring.

Modbus RTU Message Structure: PDU & ADU

PDU (Protocol Data Unit)

Function Code + Data

- **Function Code:** Specifies the action (for example, read register).
- **Data Field:** Contains parameters or relevant data for the action.



ADU (Application Data Unit)

Address + PDU + CRC

- **Address Field:** Directs the message to the correct device.
- **CRC:** Ensures data integrity.



Concept: The PDU defines the action, while the ADU adds addressing and error checking for reliable delivery.

Modbus RTU Message Framing

Framing Overview

- **ADU (Application Data Unit):** The ADU contains the key components of a Modbus message: Address, Function Code, Data, CRC.
- **ADU Structure Recap:** The ADU is the core message structure for communication.
- **Frame:** Consists of:

Address	FC	Data	CRC
1 Byte	1 Byte	0 – 252 Bytes	2 Bytes

Byte-Level Transmission

ADU Transmission: Each byte is transmitted with additional components that ensure proper reception:

- **Start Bit:** Marks the beginning of each byte.
- **Data Bits:** Contains the core data (e.g., Address or part of the Data field).
- **Parity Bit:** Optional, used for error checking (if enabled).
- **Stop Bits:** Indicates the end of each byte.

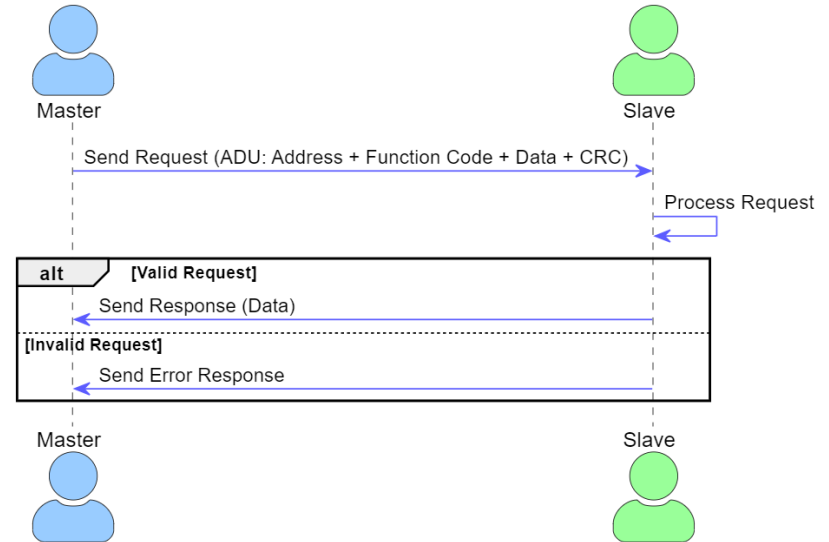


Concept Recap: The ADU is the complete logical message, while each byte of the ADU is transmitted serially with start, data, parity, and stop bits to ensure reliable communication.

Modbus RTU Master-Slave Message Flow

Master-Slave Communication

- Master initiates a request to the slave, sending an ADU (Address, Function Code, Data, CRC).
- Slave processes the request and responds accordingly.

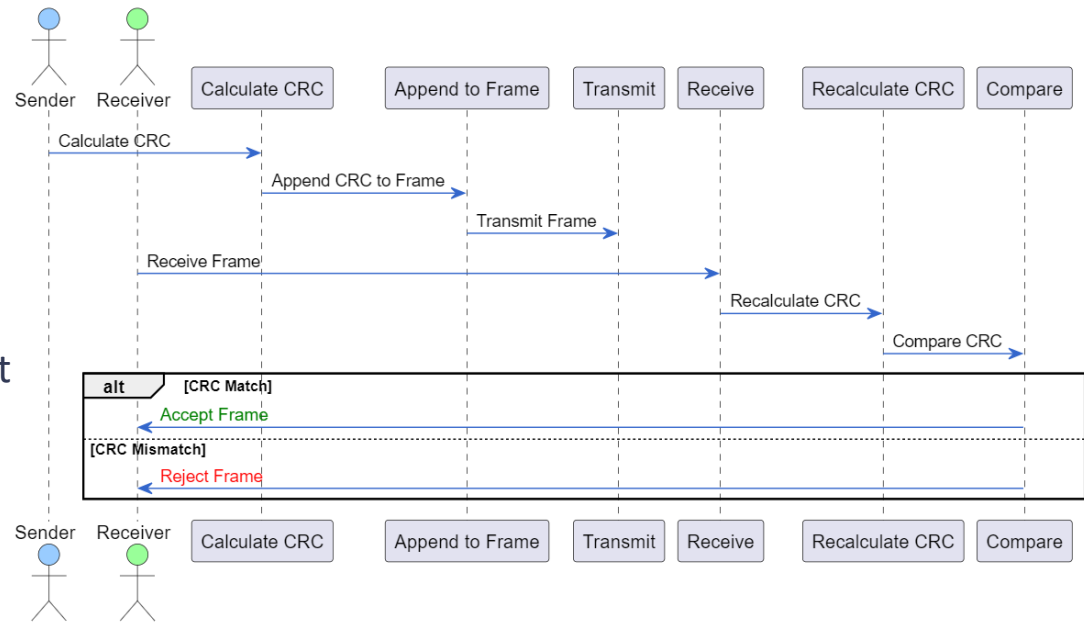


Master-Slave Message Flow

CRC in Modbus RTU: Ensuring Data Integrity

CRC Overview: Error checking mechanism. CRC (Cyclic Redundancy Check) is used to verify data integrity in Modbus RTU.

- **Placement:** Appended at the end of each frame.
- **CRC-16-MODBUS:** Uses a 16-bit polynomial for calculation, producing Low and High Bytes.



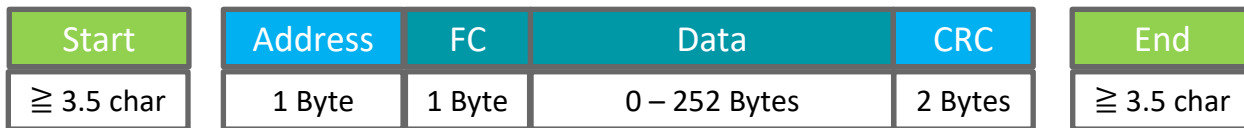
Role of CRC

- **Sender:** Calculates CRC based on frame data and appends it.
- **Receiver:** Recalculates CRC; if it matches, the data is accepted, otherwise rejected.

Timing in Modbus RTU Communication

Importance of Timing

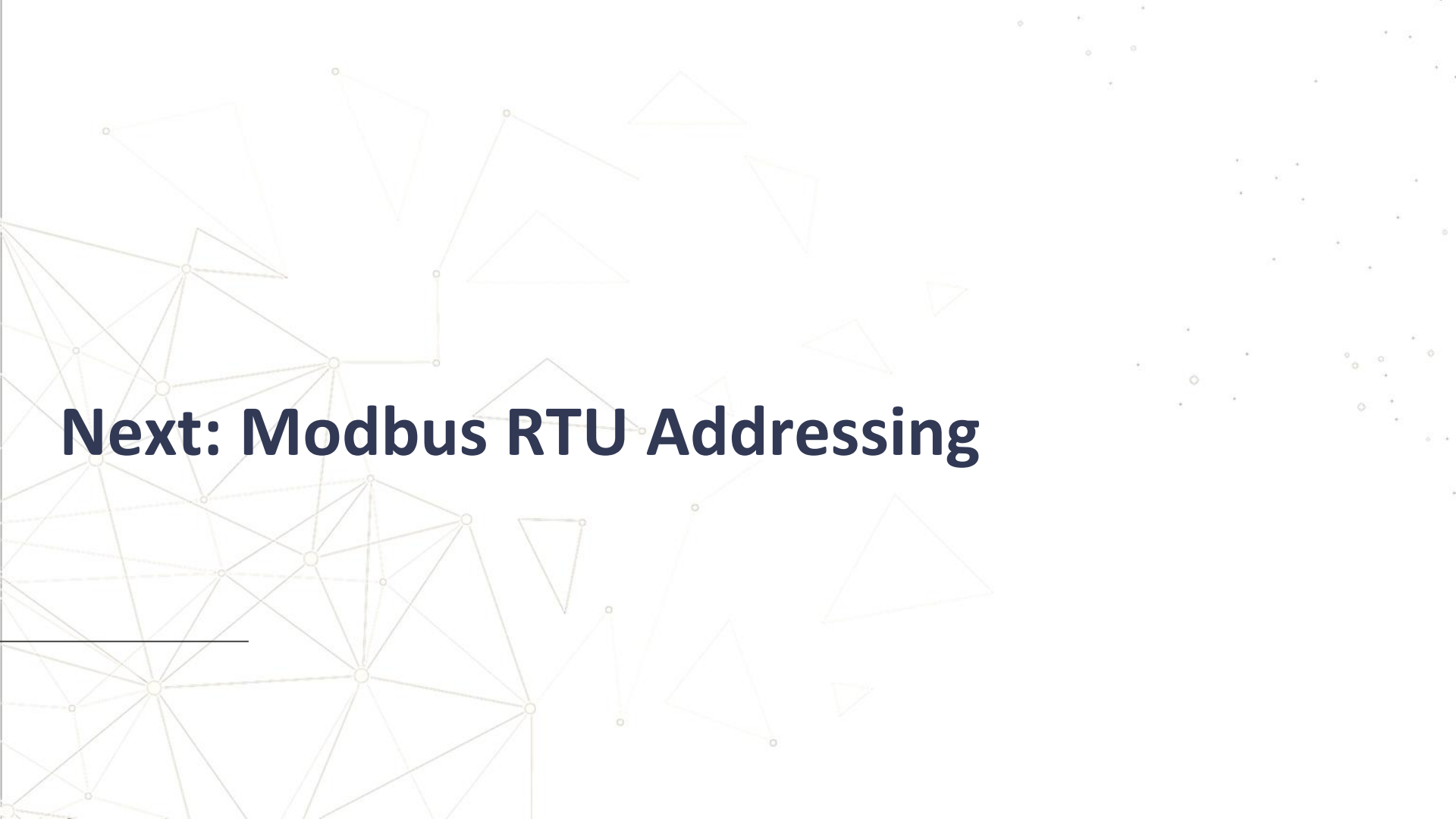
- **Silent Interval:** Before a message starts, there must be a silent interval of at least 3.5-character times to indicate the start of a new frame.



- **Inter-Character Gap:** The gap between consecutive characters must be no more than 1.5-character times to prevent the message from being discarded.

Implementation Note

- **UART IDLE Line Detection** will be used to manage silent intervals, which helps simplify handling timing requirements. However, for stricter timing, like in industrial-grade implementations, dedicated timers should be considered.

The background features a complex network of thin, light-colored lines and small circles, forming a web-like structure. Overlaid on this are several translucent triangles of various sizes and orientations, some of which are filled with a light yellow or green color. The overall aesthetic is technical and modern.

Next: Modbus RTU Addressing
